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Large Language Models, Small Labor Market Effects

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to labor market outcomes.

To examine these outcomes, we link our surveys to administrative records on monthly earnings, hours worked, and occupations through December 2024—two years after the launch of ChatGPT. Using a difference-in-differences framework, we compare adopters and non-adopters before and after the arrival of AI chatbots, estimating impacts at both the worker and workplace levels.

We begin by analyzing workers who use AI chatbots. If these tools enhance individual productivity, a natural question is whether such gains translate into higher earnings. Our first key finding is that AI chatbots have had minimal overall impact on workers who use them. Difference-in-differences estimates for earnings, recorded hours, and wages are all precisely estimated zeros, with confidence intervals ruling out average effects larger than 2%. These null results hold even for workers who use the tools daily, report substantial productivity gains, perform new AI-related tasks, and work in firms that encourage and invest in chatbots. Moreover, we find no significant effects in any of our 11 occupations, including those with flexible, decentralized wage-setting.

While chatbots have not led to earnings gains for users, we find a clear link to occupational mobility: adopters are more likely to have transitioned into new occupations after the arrival of AI chatbots. This aligns with experimental evidence that chatbots are especially beneficial for workers with less prior expertise (Brynjolfsson, Li and Raymond, 2025; Noy and Zhang, 2023), and supports the view that Generative AI helps workers extend into new occupations (Autor, 2024). However, the effects are modest—amounting to 4% of full-time equivalent employment—and do not result in net gains in hours or earnings.

Next, we zoom out to the workplace level and examine whether employers that encourage and invest in AI chatbots have experienced differential changes in labor demand. If chatbots substitute for labor, they could lead to slower hiring or trigger layoffs; if they enhance productivity, firms might expand hiring. Yet we find no such differences: workplaces that encourage chatbot use show no differential changes in employment or

E.2 Results

E.2.1 Worker Earnings

Figure E.11 shows that the null results for adopter earnings hold separately for workers' hourly wages, intensive-margin hours, and extensive-margin employment.

Figure E.12 adds non-adopters in encouraged workplaces to the main earnings plot (Figure 2.(a)), showing they have also fared similarly to their counterparts in non-encouraged workplaces following the introduction of AI chatbots.

Figure E.11: Have Adopting Workers Experienced Earnings Gains?
(Log Earnings Relative to Non-Encouraged Non-Adopters)



Notes: This figure presents the differential labor market outcomes of AI chatbot adopters relative to non-adopters, indexed to the launch of ChatGPT in November 2022. Effects are estimated separately for adopters whose employers encourage AI chatbot use (“Encouraged”) and those without encouragement (“Non-Encouraged”). Each group is compared to non-encouraged non-adopters. Estimates are based on the dynamic difference-in-differences specification in Equation (1). Shaded areas represent 95% confidence intervals. *Sample:* All completed responses from the 2024 survey linked to registry data.